2019 Ph H2 Q6

Section: Our Dynamic Universe

Topic: The Expanding Universe

Light from a distant galaxy is observed at Earth. The galaxy is moving away from Earth, causing the wavelength of light to be increased.

(a)
State the name of this effect

✅ The effect is called the **redshift**.

(b)
Explain how this observation provides evidence for the Big Bang theory

✅ A redshift means the light's wavelength is stretched, implying the galaxy is **moving away** from us.
Since **most galaxies are redshifted**, it suggests the universe is **expanding in all directions**.
An expanding universe implies that, in the past, all matter was concentrated in a very small space — which is the basis of the **Big Bang theory**.

(c)
A line in the spectrum of light from a distant galaxy has a wavelength of $5.5 \times 10^{-7}\,\mathrm{m}$.
The same line in a laboratory has a wavelength of $4.5 \times 10^{-7}\,\mathrm{m}$.
Calculate the redshift z.

Use the redshift formula:

$$z = \frac{\lambda_{\text{observed}} - \lambda_{\text{emitted}}}{\lambda_{\text{emitted}}} = \frac{5.5 \times 10^{-7} - 4.5 \times 10^{-7}}{4.5 \times 10^{-7}} = \frac{1.0 \times 10^{-7}}{4.5 \times 10^{-7}} = 0.22$$

(d)
Calculate the recessional velocity of the galaxy

Use:
$$z = \frac{v}{c} \Rightarrow v = zc = 0.22 \times 3.0 \times 10^8 = 6.6 \times 10^7\,\mathrm{m\,s^{-1}}$$

🧠 Revision Tips

- **Redshift** occurs when galaxies move **away** — spectral lines shift to longer wavelengths
- Use:
$$z = \frac{\Delta\lambda}{\lambda} = \frac{v}{c}$$
- Redshift gives direct evidence for an **expanding universe**
- The greater the redshift, the **faster** the galaxy is receding — supports Hubble's Law